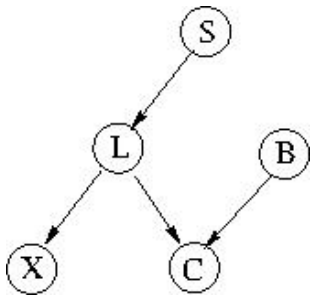
FIT3080 Artificial Intelligence
Solutions to Practice Questions for Final Exam – Bayesian Networks

1. [**Bayesian Reasoning**] Consider the following Bayesian network for another version of the medical diagnosis example, where *B=Bronchitis, S=Smoker, C=Cough, X=Positive X-ray* and *L=Lung cancer*, and all nodes are Boolean. Suppose that the prior for a patient being a smoker is 0.25, and the prior for the patient having bronchitis is 0.05. List the pairs of nodes that are conditionally independent in the following situations, and explain why.



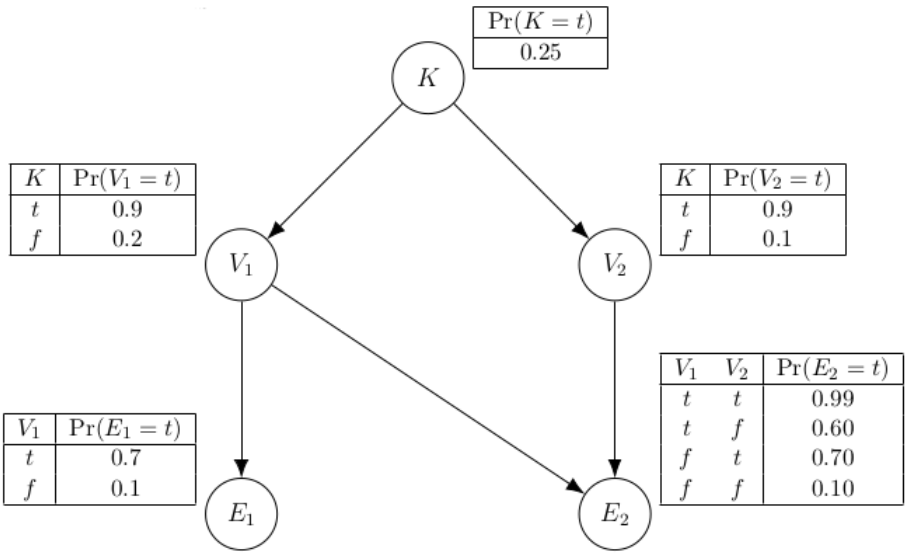
   (a) There is no evidence for any of the nodes.
   SOLUTION: L and B (C is common effect), S and B (C is common effect between L and B), X and B (C is common effect between L and B).

   (b) The lung cancer node (L) is set to true (and there is no other evidence).
   SOLUTION: S and X (chain), S and C (chain), S and B (chain to C), X and C (common cause), X and B (common cause with C).

   (c) The smoker node (S) is set to true (and there is no other evidence).
   SOLUTION: L and B (C is common effect), X and B (C is common effect between L and B). Note that S and B are not mentioned because S is an evidence node, and it is no longer a variable.

   (d) The cough node (C) is set to true (and there is no other evidence).
   SOLUTION: There are no conditional independencies in this situation.

2. [**Bayesian Networks**] Consider the following Bayesian Network. All variables are Boolean, so they can take either True (t) or False (f) as their values.



Calculate the value of $Pr(K = t, V_1 = f, V_2 = t, E_1 = f, E_2 = t)$. Show the formula and the instantiated values in the formula.

SOLUTION:

$$
\begin{aligned}
&Pr(K = t, V_1 = f, V_2 = t, E_1 = f, E_2 = t) = \\
&\quad \Pr(E_1 = f|V_1 = f) \times \Pr(E_2 = t|V_1 = f, V_2 = t) \times \Pr(V_1 = f|K = t) \times \\
&= \quad \Pr(V_2 = t|K = t) \times \Pr(K = t) \\
&= \quad (1 - 0.1) \times 0.7 \times (1 - 0.9) \times 0.9 \times 0.25 = 0.014175
\end{aligned}
$$

3. **[Bayesian Networks]** Consider the following scenario about computer systems:

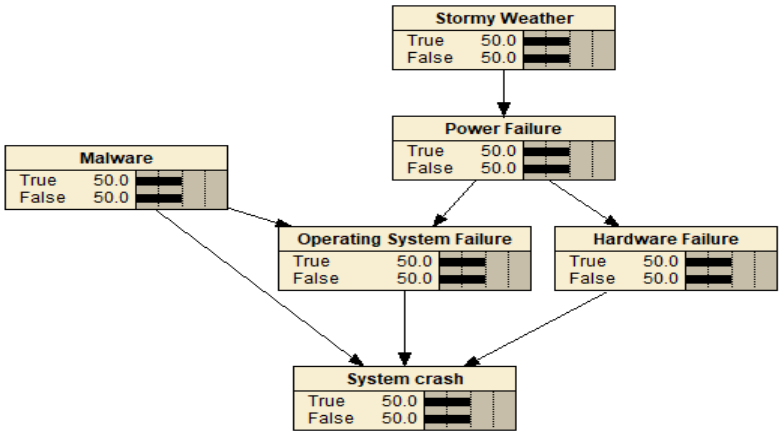
> *A* **system crash** *(***SC***) may be caused by an* **operating system failure** *(***OSF***), a* **hardware failure** *(***HF***) and/or* **malware** *(***M***). An* **OSF** *in turn may be caused by* **M** *and/or by a* **power failure** *(***PF***), which may be caused by* **stormy weather** *(***SW***). Finally, a* **PF** *may cause a* **HF**.

(a) Design a BN that represents this situation, including the conditional probability formulas for each node, e.g., $\Pr(X|Y,Z)$.

Notes:

- The boldfaced concepts should be represented in the BN.
- All the variables are binary (they have two values).
- You don't need to specify the actual probabilities in the CPTs.

SOLUTION:



The CPT probabilities are:
$\Pr(SW)$, $\Pr(M)$, $\Pr(PF|SW)$, $\Pr(OSF|M,PF)$, $\Pr(HF|PF)$, $\Pr(SC|M,OSF,HF)$.

(b) Based on the given scenario and the semantics of the nodes, could M, OSF and HF be in an Explain Away relation? Justify your answer using the features required for an Explain Away relation to hold.

SOLUTION:

The features of an explain-away relation are:

- V structure
- Parent nodes are possible causes of child node
- When child node is instantiated, the probabilities of the parent nodes increases.
- When a parent node is instantiated, the probabilities of the other parent node(s) decrease.

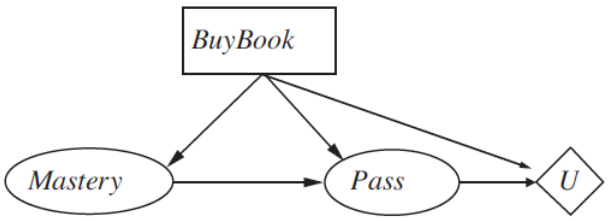
SC has three parent nodes: M, OSF and HF.

- Observing OSF would reduce the probability of HF and viceversa. So, they are in an explain-away relation.
- Similarly, observing M would reduce the probability of HF and viceversa. So, they are also in an explain-away relation.
- However, M and OSF are not in an explain-away relation, because M is a possible cause of OSF.

4. **[Bayesian Decision Networks]** Consider a student who has the choice to buy or not buy a textbook for a course. We'll model this as a decision problem with one Boolean decision Node, B, indicating whether the student chooses to buy the book, and two Boolean chance nodes, M, indicating whether the student has mastered the material in the book, and P, indicating whether the student passes the course. Of course, there is also an *additive* utility node, U. A certain student, Sam, has a utility function: 0 for not buying the book and -\$100 for buying it; and \$2000 for passing the course and 0 for not passing. In this case, the additive utility for passing and buying the book is $2000 - 100$.

   Sam's conditional probability estimates are as follows:
   $Pr(p|b,m) = 0.9$ $\quad Pr(m|b) = 0.9$
   $Pr(p|b,\neg m) = 0.5$ $\quad Pr(m|\neg b) = 0.7$
   $Pr(p|\neg b,m) = 0.8$
   $Pr(p|\neg b,\neg m) = 0.3$
   You might think that P would be independent of B given M, but this course has an open-book final – so having the book helps.

   (a) Draw the decision network for this problem.
   SOLUTION:



   (b) Compute the expected utility of buying the book and of not buying it.
   SOLUTION: For each of $B = b$ and $B = \neg b$, we compute $Pr(p|B)$ and thus $Pr(\neg p|B)$ by marginalizing out M, then use this to compute the expected utility.

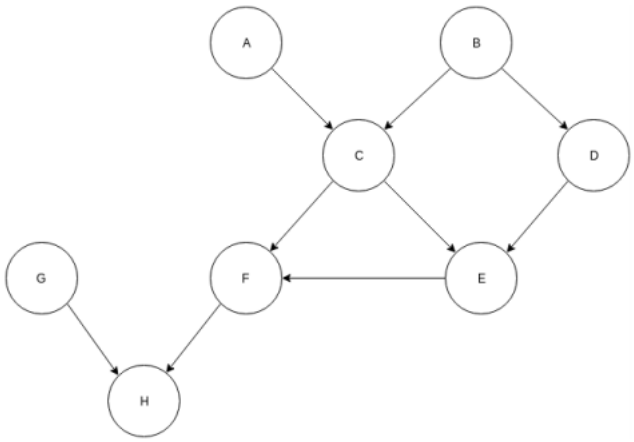
$$\begin{aligned} Pr(p|b) &= \sum_m Pr(p|b,m) \times Pr(m|b) \\ &= 0.9 \times 0.9 + 0.5 \times 0.1 \\ &= 0.86 \\ Pr(p|\neg b) &= \sum_m Pr(p|\neg b,m) \times Pr(m|\neg b) \\ &= 0.8 \times 0.7 + 0.3 \times 0.3 \\ &= 0.65 \end{aligned}$$

   The expected utilities are thus:

$$\begin{aligned} EU[b] &= \sum_p Pr(p|b) \times U(p,b) \\ &= 0.86 \times (2000 - 100) + 0.14 \times (-100) \\ &= 1620 \\ EU[\neg b] &= \sum_p Pr(p|\neg b) \times U(p,\neg b) \\ &= 0.65 \times 2000 + 0.35 \times 0 \\ &= 1300 \end{aligned}$$

   (c) What should Sam do?
   SOLUTION: Buy the book, Sam.

5. [**D-separation**] According to this Figure, determine whether the following claims to be True or False and justify your answer.



(a) $B \perp G|A$
SOLUTION: True, they are independent. Without knowing H (common effect), the information of B cannot propagate to G. It does not matter whether A is known.

(b) $C \perp D|F$
SOLUTION: False, they are not independent. Knowing F allows the information in C to propagate through E (common effect), then propagate to D (common effect).

(c) $C \perp D|A$
SOLUTION: False, they are not independent. Without knowing B (common cause), the information in C can propagate to D. Note that if there was evidence for E (common effect), information could propagate from C to D through E. However, in this situation we have no evidence for E, hence this path is blocked. Also note that it does not matter whether A is known.

(d) $H \perp B|C, F$
SOLUTION: True, they are independent. Knowing C and F (chain rule) blocks the information between H and B.